Ansible Setup

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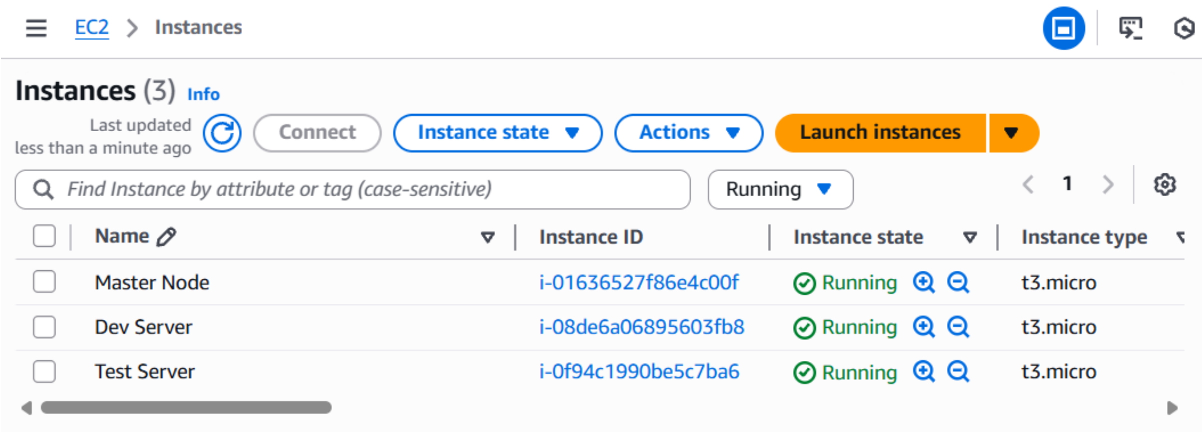
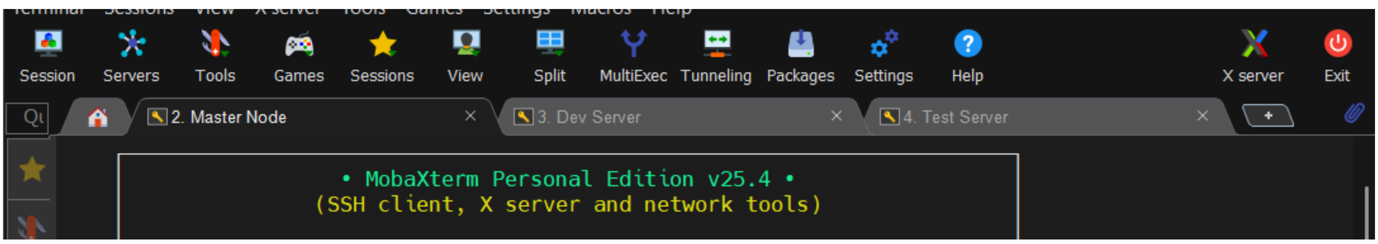
1. To set up a fully functional Ansible environment with one Control Node and multiple Worker Nodes, enabling automated configuration management and deployment.

   - Launch 3 EC2 instances: one as the **Control Node** (Master) and two as **Worker Nodes** (Dev Server and Test Server).



   - Connect to all three nodes using **MobaXterm**.



- Set the hostname for all nodes and switch to root using sudo -i.

```
Multi-execution mode: commands are typed to all terminals    Multi-paste    Exit multi-execution mode

[root@ip-172-31-5-185 ec2-user]# hostnamectl set-hostname
master
[root@ip-172-31-5-185 ec2-user]# sudo -i
[root@master ~]#

Exclude "Master Node" from MultiExec mode

[root@ip-172-31-6-240 ec2-user]# hostnamectl set-hostname
dev-server
[root@ip-172-31-6-240 ec2-user]# sudo -i
[root@dev-server ~]#

Exclude "Dev Server" from MultiExec mode

[root@ip-172-31-15-145 ec2-user]# hostnamectl set-hostname
 test-server
[root@ip-172-31-15-145 ec2-user]# sudo -i
[root@test-server ~]#
```

- Install **Ansible** on the Control Node.

```
[ec2-user@master ~]$ ansible --version
ansible [core 2.15.13]
  config file = None
  configured module search path = ['/home/ec2-user/.ansible/plugins/modules', '/usr/share/ansible/
plugins/modules']
  ansible python module location = /usr/local/lib/python3.9/site-packages/ansible
  ansible collection location = /home/ec2-user/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/bin/ansible
  python version = 3.9.25 (main, Dec 10 2025, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (
/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
```

- Edit the SSH configuration file /etc/ssh/sshd_config:
  - Set PermitRootLogin yes (line 38)
  - Set PasswordAuthentication yes (line 63)

  This allows commands to be run on other servers as the root use

```
2. Master Node    3. Dev Server

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
```

2. Master Node | 3. Dev Server | 4.

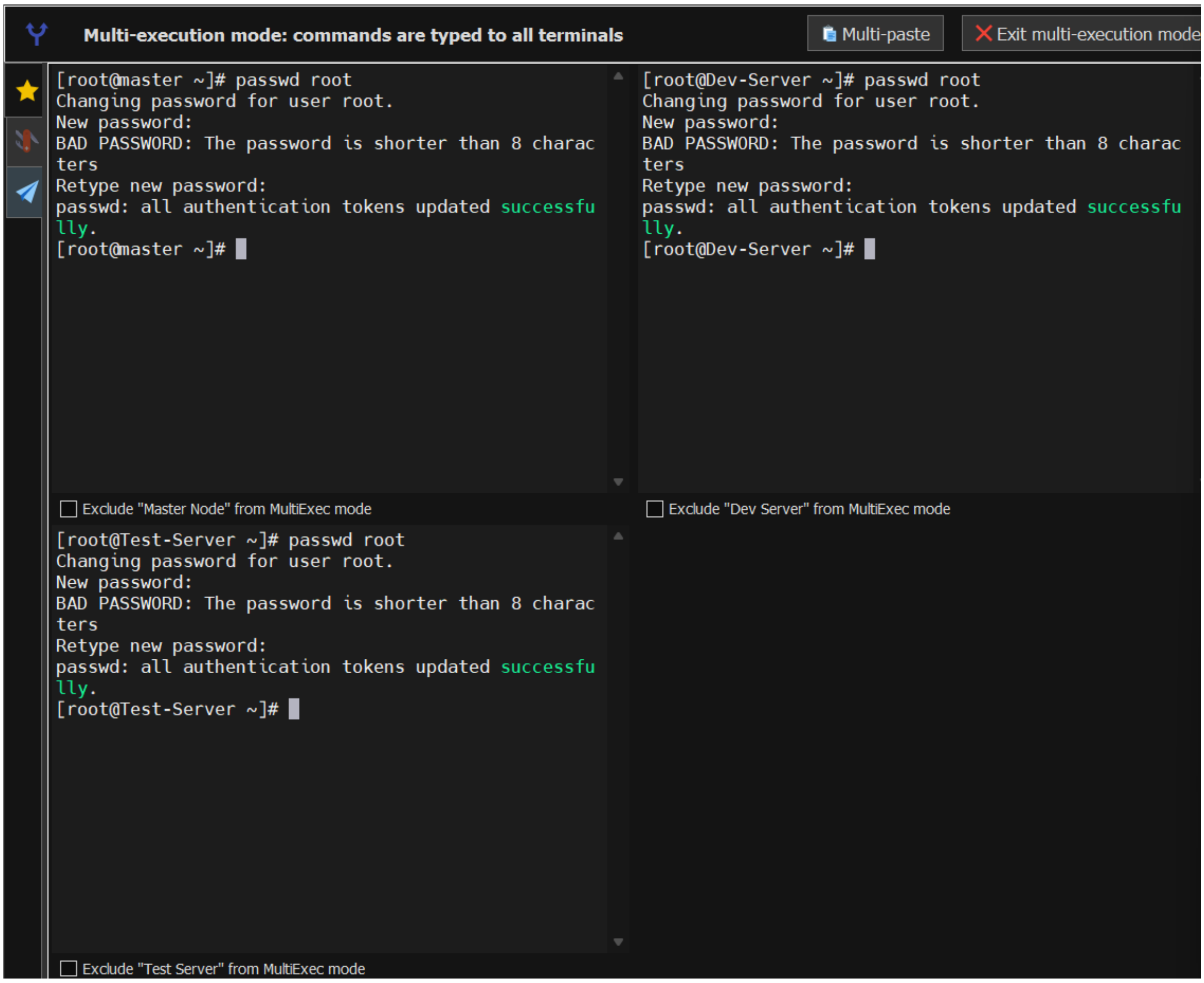
```
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes
```

- Apply the SSH configuration changes to **all nodes**.
- Restart the SSH service on all nodes for changes to take effect

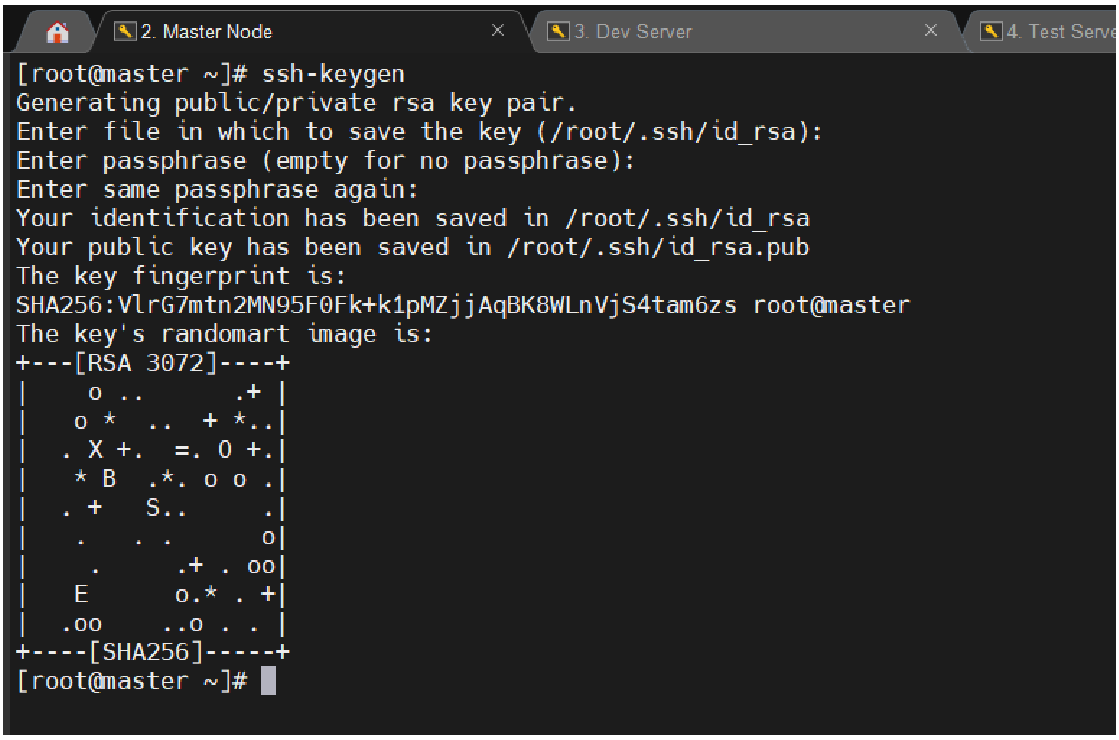
2. Master Node | 3. Dev Server | 4. Test Server

```
[root@master ~]# vi /etc/ssh/sshd_config
[root@master ~]# systemctl restart sshd
[root@master ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-02-04 13:56:55 UTC; 33s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
   Main PID: 16717 (sshd)
      Tasks: 1 (limit: 1067)
     Memory: 1.4M
        CPU: 9ms
     CGroup: /system.slice/sshd.service
             └─16717 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 04 13:56:55 master systemd[1]: Starting sshd.service - OpenSSH server daemon...
Feb 04 13:56:55 master sshd[16717]: Server listening on 0.0.0.0 port 22.
Feb 04 13:56:55 master sshd[16717]: Server listening on :: port 22.
Feb 04 13:56:55 master systemd[1]: Started sshd.service - OpenSSH server daemon.
[root@master ~]#
```

- Set passwords for all nodes to enable root access.



- Generate an **SSH key** on the Control Node using ssh-keygen



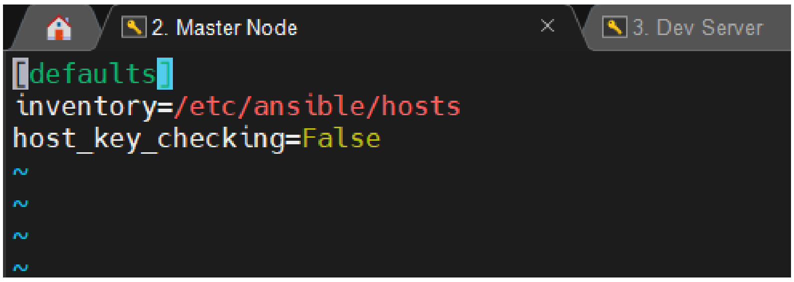
- Using the ssh key we establish the connection between master nodes and worker nodes and login worker node from the master node
- Use the **private IPs** of worker nodes for Ansible connections.

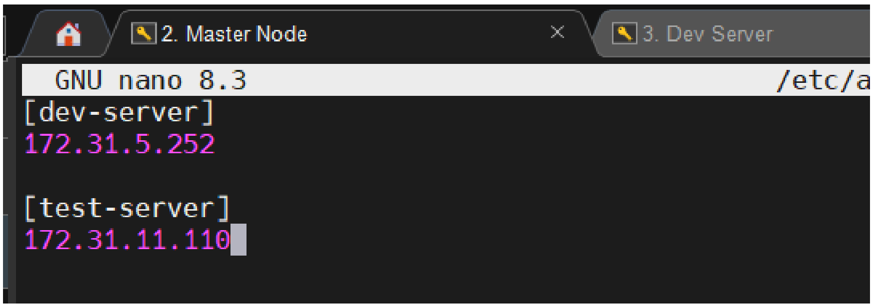
```
[root@master ~]# ssh-copy-id root@172.31.5.252
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already ins
talled
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the
new keys
root@172.31.5.252's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@172.31.5.252'"
and check to make sure that only the key(s) you wanted were added.

[root@master ~]# ssh 172.31.5.252
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Wed Feb  4 13:17:34 2026
[root@Dev-Server ~]#
```

```
2. Master Node    3. Dev Server    4. Test Server
[root@master ~]# ssh-copy-id root@172.31.11.110
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host '172.31.11.110 (172.31.11.110)' can't be established.
ED25519 key fingerprint is SHA256:82y8MkfeUw0a/tYKWLbEEM92BmNUNpw9DpN/+jTblXI.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already ins
talled
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the
new keys
root@172.31.11.110's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'root@172.31.11.110'"
and check to make sure that only the key(s) you wanted were added.

[root@master ~]# ssh 172.31.11.110
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Wed Feb  4 13:17:34 2026
[root@Test-Server ~]#
```

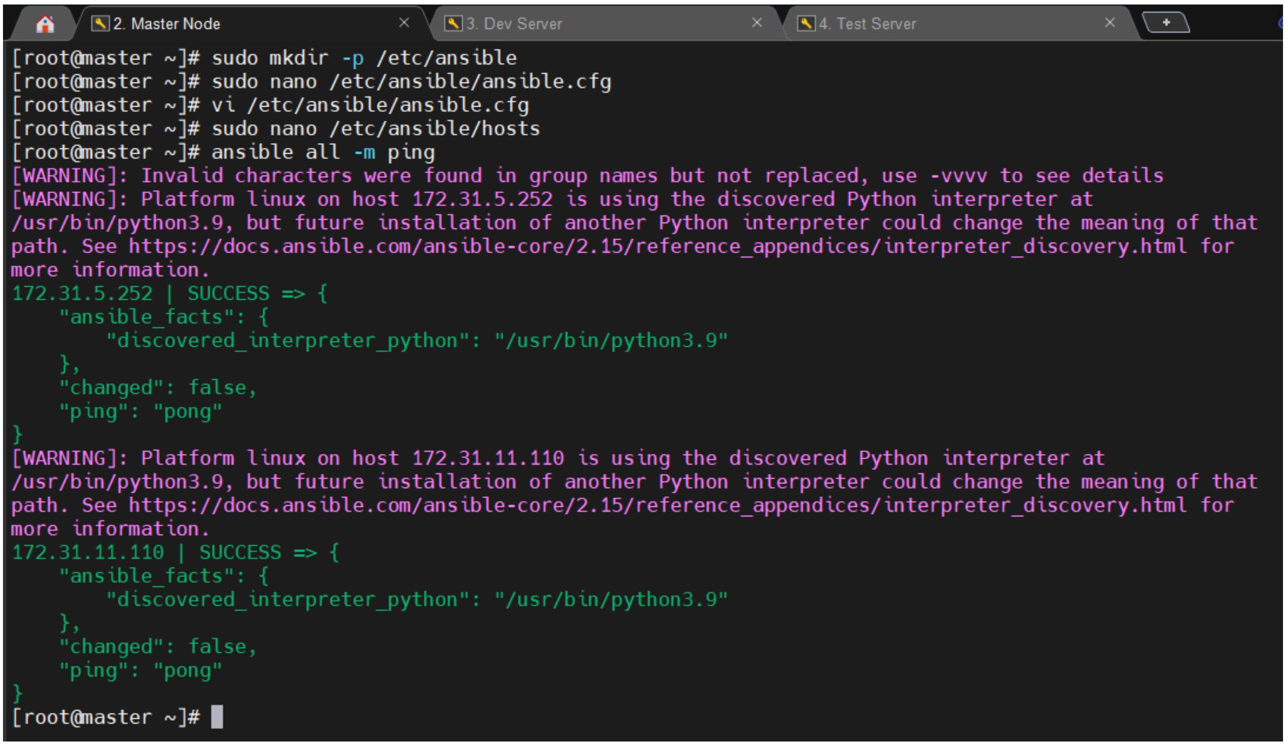
- Configure the **hosts inventory file** on the Control Node to define the Worker Nodes. Inventory tells Ansible which servers to manage.

```
2. Master Node    3. Dev Server
[root@master ~]# sudo mkdir -p /etc/ansible
[root@master ~]# sudo nano /etc/ansible/ansible.cfg
```





- Test the connection from Control Node to Worker Nodes using: *ansible all -m ping* Expected output: {"ping": "pong"}



- ➢ Control Node can now communicate with Worker Nodes.
- ➢ Ansible inventory is set up.
- ➢ Environment is ready for running playbooks and roles.